

Q1 - 24 June - Shift 1

Sodium light of wavelengths 650 nm and 655 nm is used to study diffraction at a single slit of aperture 0.5 mm. The distance between the slit and the screen is 2.0 m. The separation between the positions of the first maxima of diffraction pattern obtained in the two cases is _____ $\times 10^{-5}$ m.

Space for your notes:

Q2 - 25 June - Shift 1

The two light beams having intensities I and $9I$ interfere to produce a fringe pattern on a screen. The phase difference between the beams is $\frac{\pi}{2}$ at point P and π at point Q. Then the difference between the resultant intensities at P and Q will be :

Space for your notes:

- (A) $2I$ (B) $6I$
(C) $5I$ (D) $7I$

Q3 - 25 June - Shift 2

The interference pattern is obtained with two coherent light sources of intensity ratio 4 : 1. And the ratio $\frac{I_{\max} + I_{\min}}{I_{\max} - I_{\min}}$ is $\frac{5}{x}$. Then, the value of x will be equal to :

Space for your notes:

- (A) 3 (B) 4
(C) 2 (D) 1

Q4 - 26 June - Shift 1

Questions

MathonGo

In free space, an electromagnetic wave of 3 GHz of 3 GHz frequency strikes over the edge of an object of size $\frac{\lambda}{100}$, where λ is the wavelength of the wave in free space. The phenomenon, which happens there will be:

- (A) Reflection (B) Refraction
(C) Diffraction (D) Scattering

*Space for your notes:***Q5 - 26 June - Shift 2**

For a specific wavelength 670 nm of light coming from a galaxy moving with velocity v , the observed wavelength is 670.7 nm.

The value of v is :

- (A) $3 \times 10^8 \text{ ms}^{-1}$ (B) $3 \times 10^{10} \text{ ms}^{-1}$
(C) $3.13 \times 10^5 \text{ ms}^{-1}$ (D) $4.48 \times 10^5 \text{ ms}^{-1}$

*Space for your notes:***Q6 - 27 June - Shift 1**

In Young's double slit experiment the two slits are 0.6 mm distance apart. Interference pattern is observed on a screen at a distance 80 cm from the slits. The first dark fringe is observed on the screen directly opposite to one of the slits. The wavelength of light will be _____ nm.

*Space for your notes:***Q7 - 28 June - Shift 2**

#MathBoleTohMathonGo

In young's double slit experiment performed using a monochromatic light of wavelength λ , when a glass plate ($\mu = 1.5$) of thickness $x\lambda$ is introduced in the path of the one of the interfering beams, the intensity at the position where the central maximum occurred previously remains unchanged. The value of x will be:

- (A) 3 (B) 2
(C) 1.5 (D) 0.5

Space for your notes:

Q8 - 28 June - Shift 2

In a Young's double slit experiment, an angular width of the fringe is 0.35° on a screen placed at 2 m away for particular wavelength of 450 nm. The angular width of the fringe, when whole system is immersed in a medium of refractive index $7/5$, is $\frac{1}{\alpha}$. The value of α is _____

Space for your notes:

Q9 - 29 June - Shift 1

Questions

MathonGo

Using Young's double slit experiment, a monochromatic light of wavelength $5000 \text{ \AA}$ produces fringes of fringe width 0.5 mm . If another monochromatic light of wavelength $6000 \text{ \AA}$ is used and the separation between the slits is doubled, then the new fringe width will be :

- (A) 0.5 mm (B) 1.0 mm
(C) 0.6 mm (D) 0.3 mm

*Space for your notes:***Q10 - 29 June - Shift 2**

In a double slit experiment with monochromatic light, fringes are obtained on a screen placed at some distance from the plane of slits. If the screen is moved by $5 \times 10^{-2} \text{ m}$ towards the slits, the change in fringe width is $3 \times 10^{-3} \text{ cm}$. If the distance between the slits is 1 mm , then the wavelength of the light will be _____ nm.

Space for your notes:

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Answer Key

Q1 (3)

Q2 (B)

Q3 (D)

Q4 (D)

Q5 (C)

Q6 (450)

Q7 (B)

Q8 (4)

Q9 (D)

Q10 (600)

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Q1 (3)

$$a \sin \theta = \frac{3\lambda}{2}$$

$$\frac{y}{L} = \theta = \frac{3\lambda}{2a}$$

$$L = 2\text{m}$$

$$y_1 = \frac{3\lambda_1 L}{2a}$$

$$\lambda_2 = 655 \text{ nm}$$

$$y_2 = \frac{3\lambda_2 L}{2a}$$

$$\lambda_1 = 650 \text{ nm}$$

$$a = 0.5 \text{ nm}$$

$$\Delta y = y_2 - y_1 = \frac{3(\lambda_2 - \lambda_1)L}{2a}$$

$$= \frac{3(655 - 650)}{2 \times 0.5 \times 10^{-3}} \times 2 \times 10^{-9}$$

$$= \frac{3 \times 5 \times 2}{1 \times 10^{-3}} \times 10^{-9}$$

$$= 3 \times 10^{-5}$$

Q2 (B)

$$I_p = I + 9I + 2\sqrt{I \times 9I} \cos \frac{\pi}{2}$$

$$I_p = 10I$$

$$I_Q = I + 9I + 2\sqrt{I \times 9I} \cos \pi$$

$$= 10I - 6I = 4I$$

$$\therefore I_p - I_Q = 10I - 4I = 6I$$

Q3 (D)

$$\frac{I_1}{I_2} = 4$$

$$\frac{I_{\max}}{I_{\min}} = \left[\frac{\sqrt{I_1} + \sqrt{I_2}}{\sqrt{I_1} - \sqrt{I_2}} \right]^2$$

$$\frac{I_{\max}}{I_{\min}} = \left[\frac{2\sqrt{I_2} + \sqrt{I_2}}{2\sqrt{I_2} - \sqrt{I_2}} \right]^2$$

$$\frac{I_{\max}}{I_{\min}} = 9$$

$$\frac{I_{\max} + I_{\min}}{I_{\max} - I_{\min}} = \frac{10}{8}$$

$$\frac{5}{x} = \frac{10}{8}$$

$$x = 4$$

Q4 (D)

$$\frac{a}{\lambda} = \frac{1}{100}$$

For reflection, size of obstacle must be much larger than wavelength, for diffraction size should be order of wavelength.

Since the object is of size $\frac{\lambda}{100}$, much smaller than wavelength, so scattering will occur.

Q5 (C)

$$\lambda_{\text{emitted}} = 670 \text{ nm}$$

$$\lambda_{\text{obs}} = 670.7 \text{ nm}$$

$$v = ?$$

$$c = 3 \times 10^8 \text{ m/s}$$

$$\text{If } v \ll c$$

$$\frac{\lambda_{\text{obs}} - \lambda_{\text{emitted}}}{\lambda_{\text{emitted}}} = \frac{v}{c}$$

$$\frac{670.7 - 670}{670} = \frac{v}{c}$$

$$v = 3.13 \times 10^5 \text{ m/s}$$

Q6 (450)

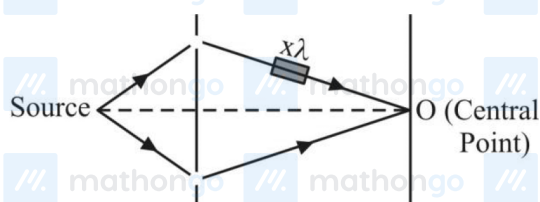
$$d = 0.6 \times 10^{-3}$$

$$D = 80 \times 10^{-2}$$

$$\text{1st Dark fringe} = \frac{D\lambda}{2d} = \frac{d}{2}, \quad \lambda = \frac{d^2}{D}$$

$$= 450 \times 10^{-9} \text{ m}$$

Q7 (B)



Path difference at O = $(\mu - 1)t$.

If the intensity at O remains (maximum)

unchanged, path difference must be $n\lambda$.

$$\Rightarrow (\mu - 1)t = n\lambda$$

$$(1.5 - 1)x\lambda = n\lambda$$

$$\Rightarrow x = 2n$$

For $n = 1$, $x = 2$

Q8 (4)

$$\beta = \frac{0.35 \times 5}{7} = 0.25$$

$$\frac{1}{\alpha} = \frac{25}{100}$$

$$\boxed{\alpha = 4}$$

Q9 (D)

$$\text{Fringe width } \beta = \frac{D\lambda}{d}$$

$$\lambda_1 = 5000 \text{ \AA}$$

$$\beta_1 = \frac{D}{d}(5000 \times 10^{-10}) = 5 \times 10^{-4} \text{ m} \dots \text{(I)}$$

$$\beta_2 = \frac{D}{(2d)}(6000 \times 10^{-10}) = x \text{ (let)} \dots \text{(II)}$$

Divide (II) & (I)

$$\frac{\beta_2}{\beta_1} = \frac{3000 \times 10^{-10}}{5000 \times 10^{-10}} = \frac{x}{5 \times 10^{-4}}$$

$$x = 3 \times 10^{-4} \text{ m or } 0.3 \text{ mm}$$

Q10 (600)

$$\beta = \frac{\lambda D}{d}$$

$$\Delta\beta = \frac{\lambda}{d} \Delta D$$

$$\lambda = \frac{\Delta\beta \cdot d}{\Delta D}$$

$$= \frac{3 \times 10^{-5} \times 1 \times 10^{-3}}{5 \times 10^{-2}}$$

$$= 60 \times 10^{-8} = 600 \times 10^{-9} \text{ m}$$

$$= 600 \text{ nm}$$